UNIVERSITY OF TWENTE.

OLFAR – Orbitting Low Frequency Antennas for Radio Astronomy

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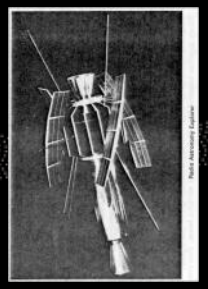
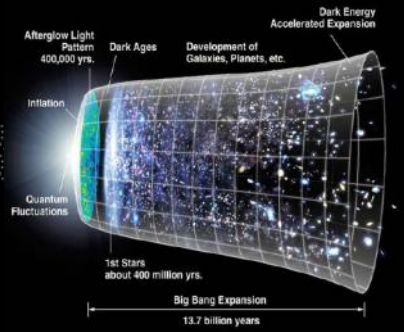
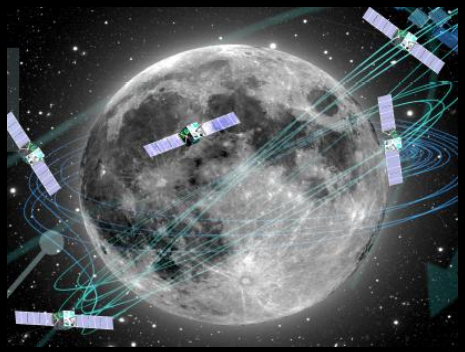






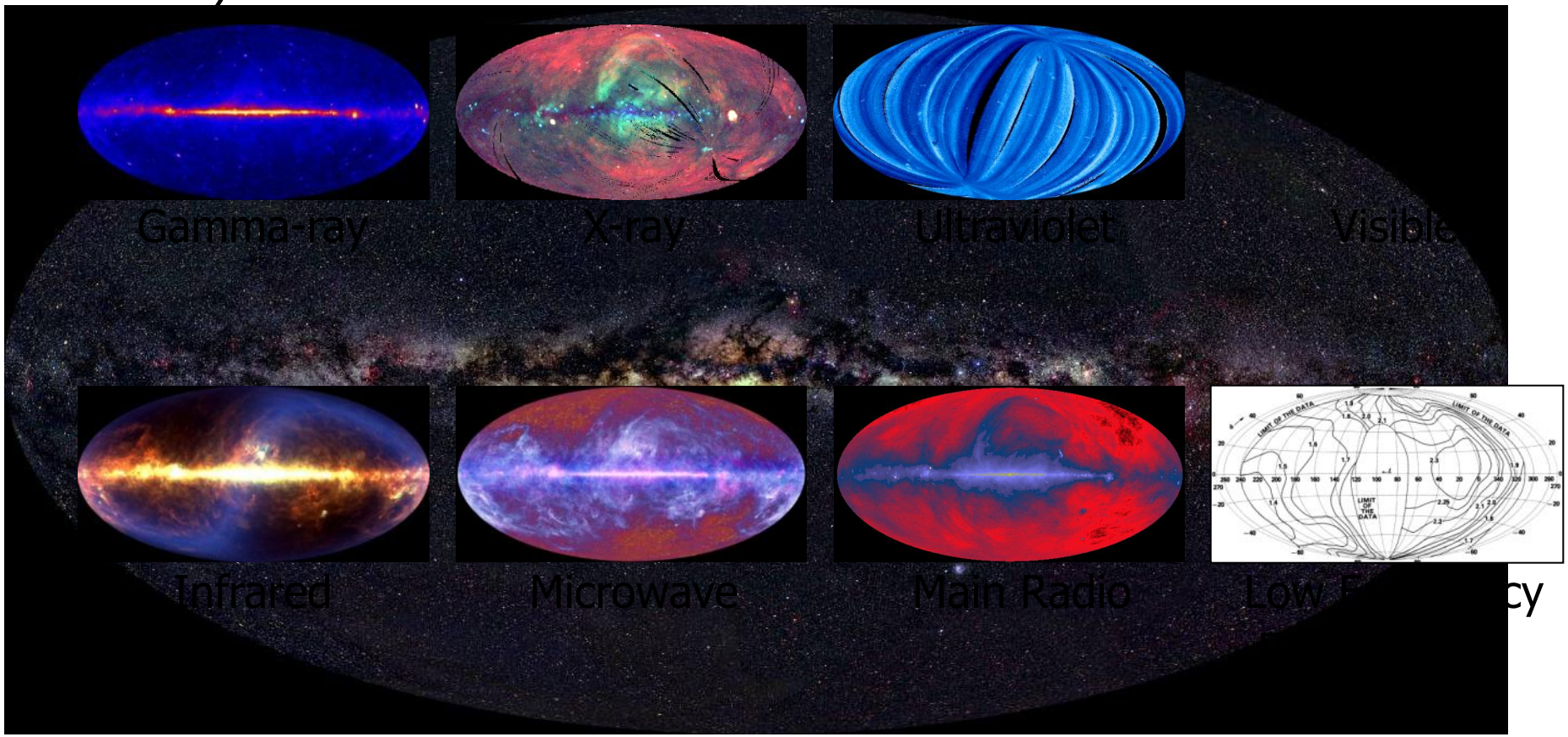
Content

Topic:
Orbitting Low Frequency Antennas for Radio Astronomy

- Background – need for a low frequency radio telescope
- Why in Space ?
- Why is it interesting ?
- Concept of OLFAR
- Requirements
- Selection of research results
  - Mission Analysis
  - Science antenna
  - Inter satellite link

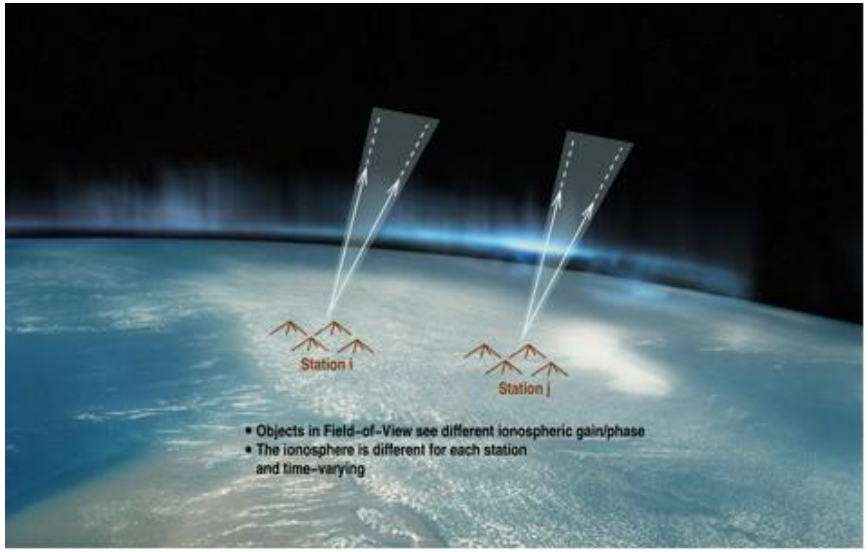
Introduction

- **Celestial sky:** has been mapped for nearly every type of electromagnetic radiation
- major exception: ultra-low frequency radiowaves (<30 MHz)



Goal: an ultra long wavelength radio telescope

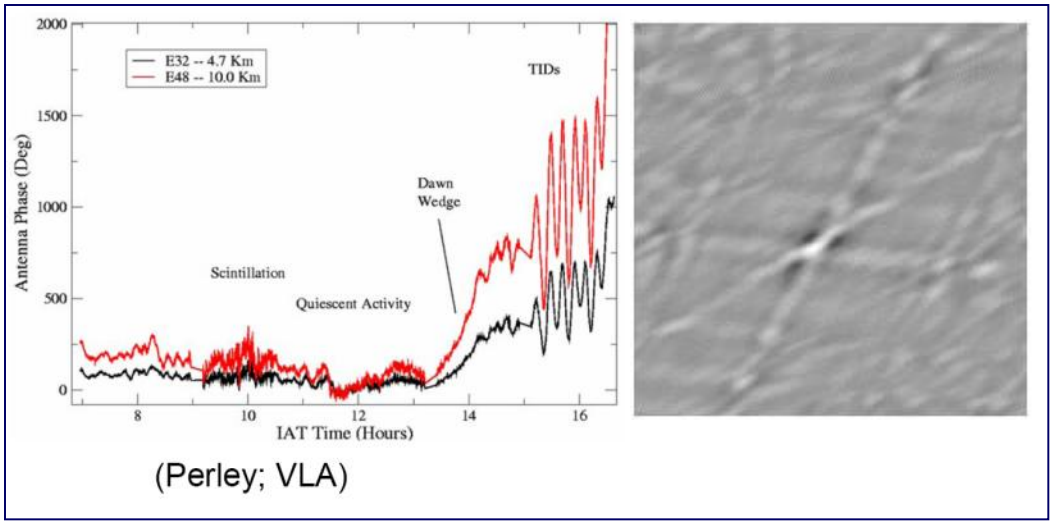
- Low frequency : below 30 MHz
- > ~30 MHz : LOFAR



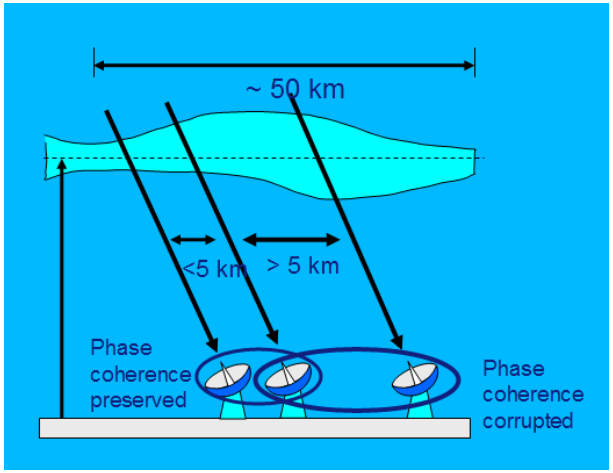
➔ Opening up the last unexplored frequency regime

Why in space: ionospheric disturbance and cut-off

- Phase coherence through ionosphere
  - Corruption of coherence of phase on baselines
- Isoplanatic Patch Problem:
  - Calibration changes as a function of position



VLA image (movie) showing ionospheric disturbances of sky image, Perley, Lazio, 74 MHz



Maximum antenna separation:
< 5 km vs. $>10^2$ km
Angular resolution at 10 MHz:
$\theta > 0.3°$ vs. $< 10^{-2}$ °

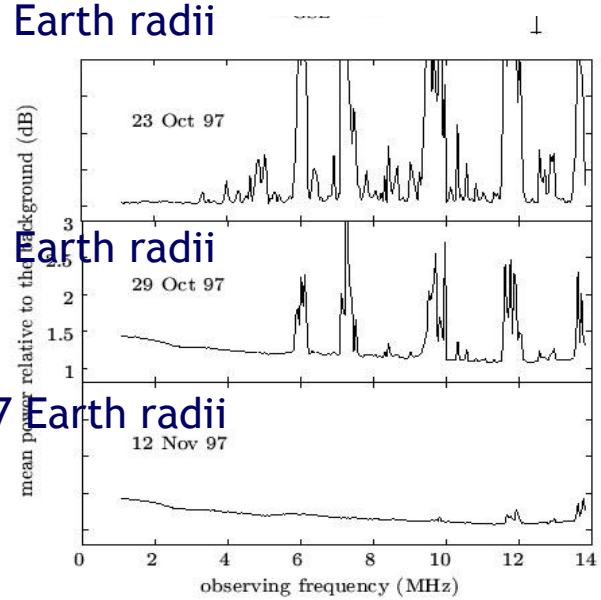
Why in space: radio frequency interference

Typical man-made interference received by the WAVES instrument on Wind

40 Earth radii

93 Earth radii

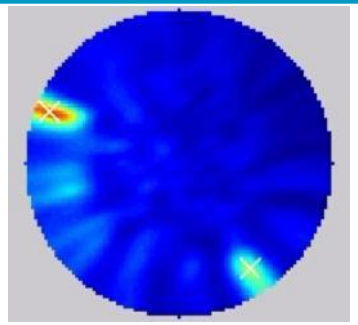
157 Earth radii



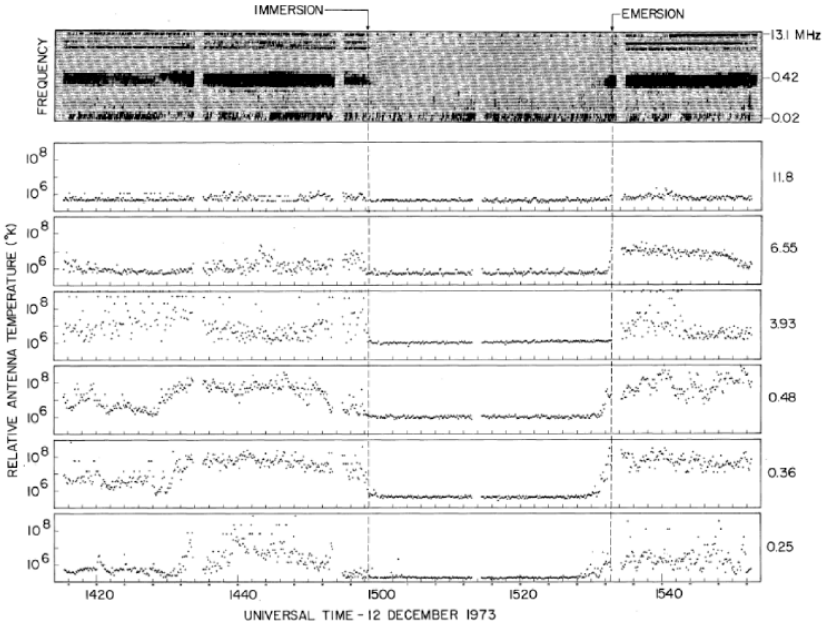
G. Woan from ESA study SCI(97)2

All-sky image (movie) of LOFAR ITS at 15.004 MHz

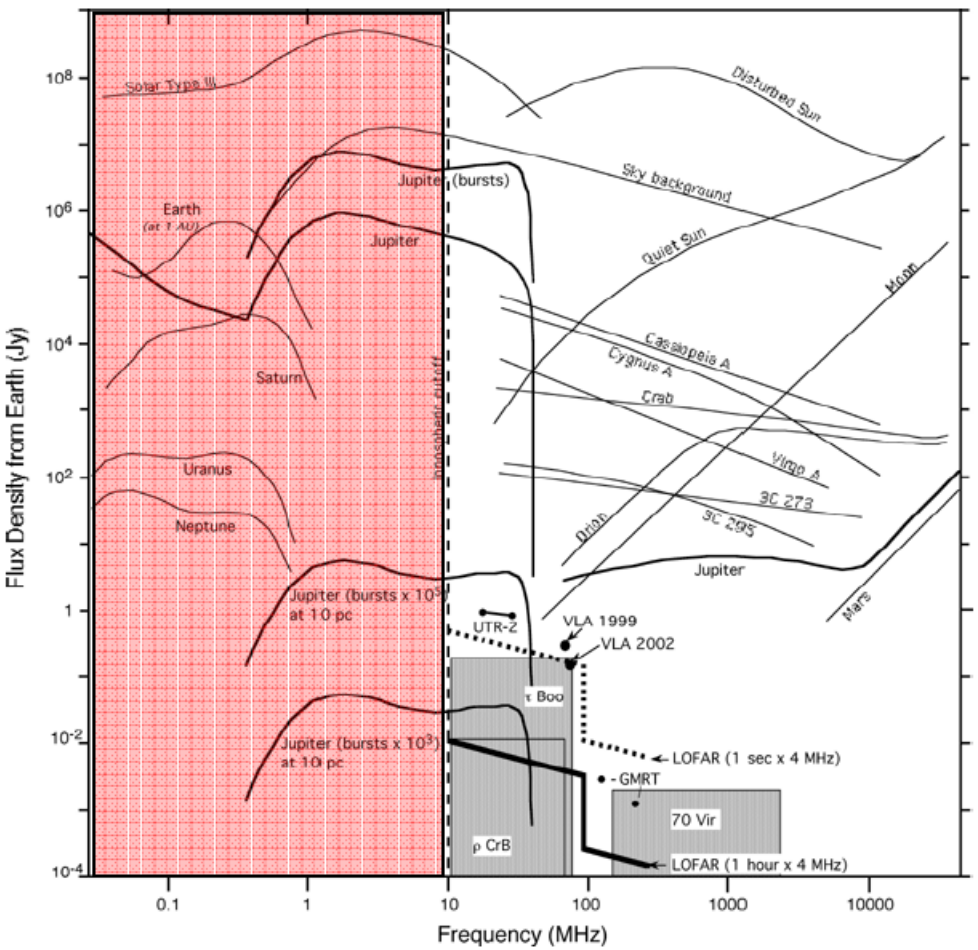
(Moren, Boonstra, 2007)



RAE B (1973) in Lunar orbit, Kaiser 1975



Why is it interesting: New frequency window

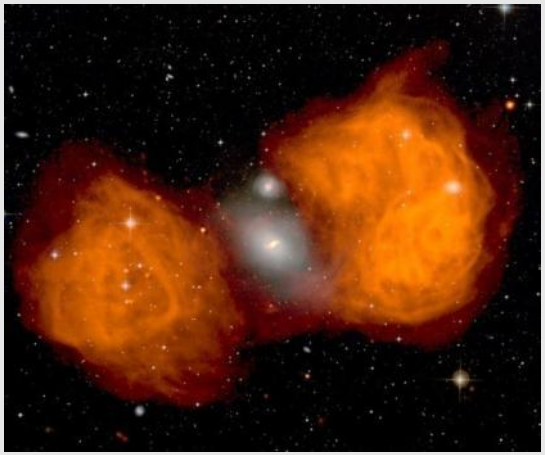


- At very low frequencies the universe is virtually unexplored
- The earth's ionosphere masks the key low frequencies

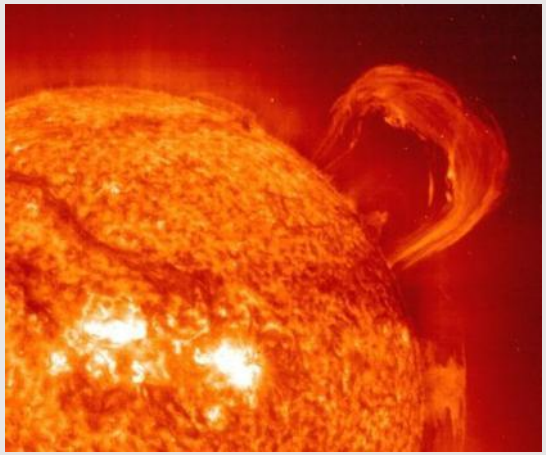
🡸 Spectra of astronomical sources.

Why is it interesting: Radio sources

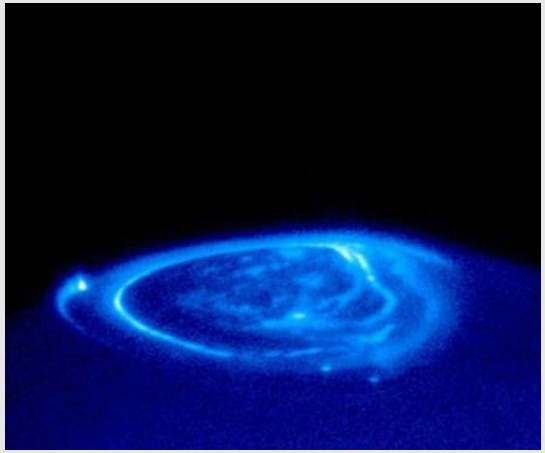
Astronomical radio sources



Radio galaxies
(static source)

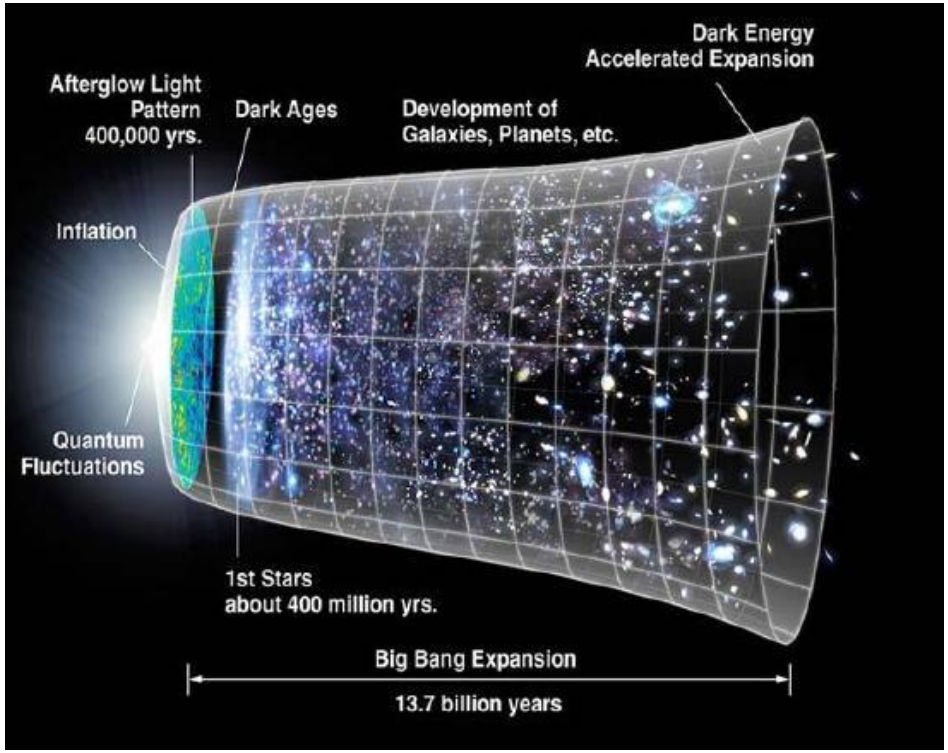


Solar eruptions
(transient source)



Aurora's of planets
(transient source)

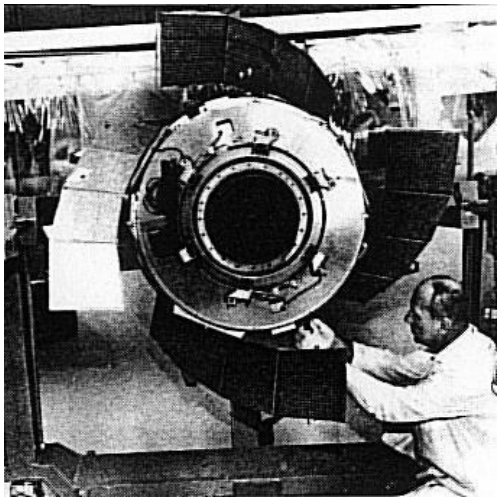
Why is it interesting: Dark ages

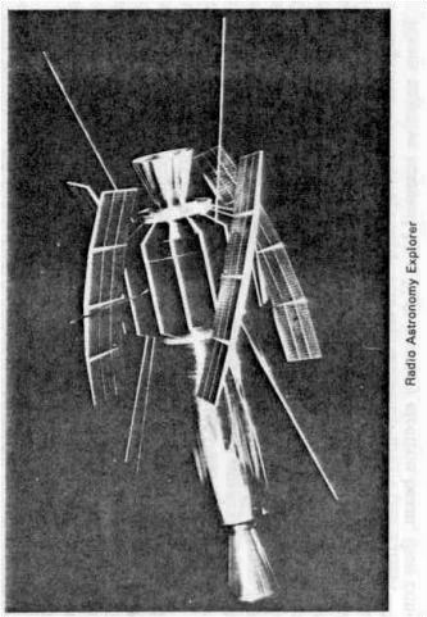


- The highly red-shifted 21 cm hydrogen line is the only known way to observe the dark ages
- At that time the universe was filled with neutral hydrogen and opaque to visible light
- Neutral Hydrogen absorbs photons and re-emit them in random directions
- During the “epoch of re-onization” new stars ionized the hydrogen and the universe became transparent again. (~400 million years after the BB)

Previous low frequency missions

- RAE-A (Explorer 38)
  - 1968 July 4
  - 190 kilogram
  - Earth orbit
- RAE- B (Explorer 49)
  - 1973 June 10
  - 328 kilogram
  - Moon orbit
  - 25 kHz to 13.1 MHz





Current Status at Long Wavelengths

Extremely poor resolution, strong diffuse Galactic emission

RAE-2 observations at
1.3 and 4.7 MHz, Novaco
& Brown 1978

slide from G. Woan

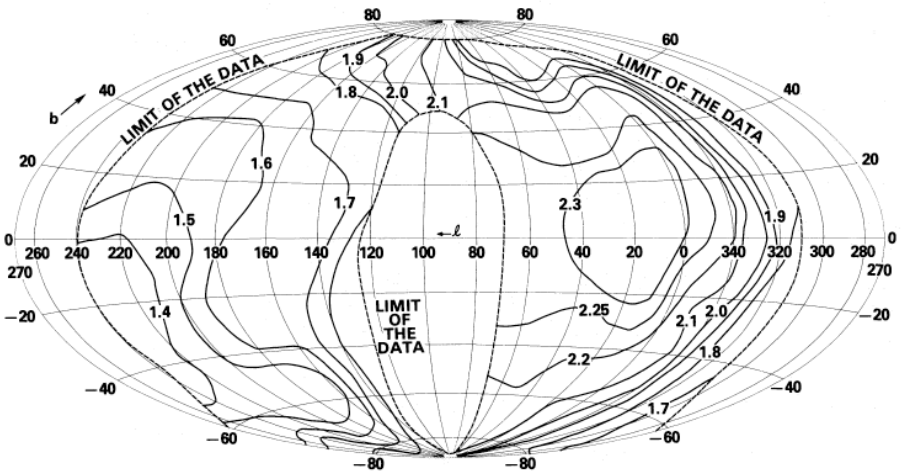


FIG. 5.—Contour map in galactic coordinates of the nonthermal emission observed by *RAE 2* at 4.70 MHz

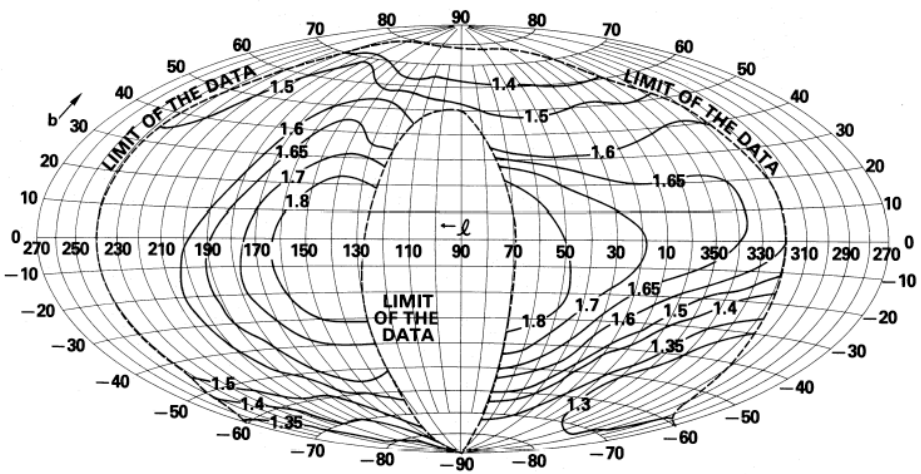
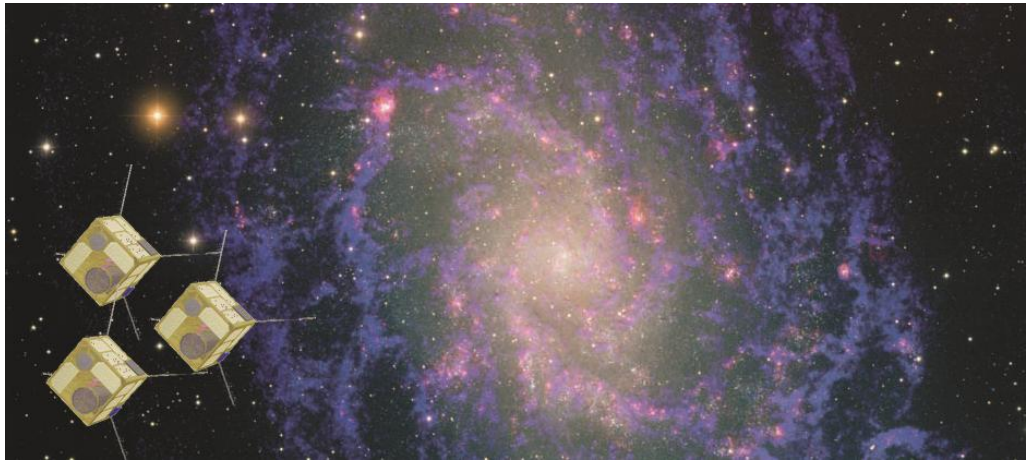


FIG. 8.—Contour map in galactic coordinates of the nonthermal emission observed by *RAE 2* at 1.31 MHz

Basic idea of OLFAR

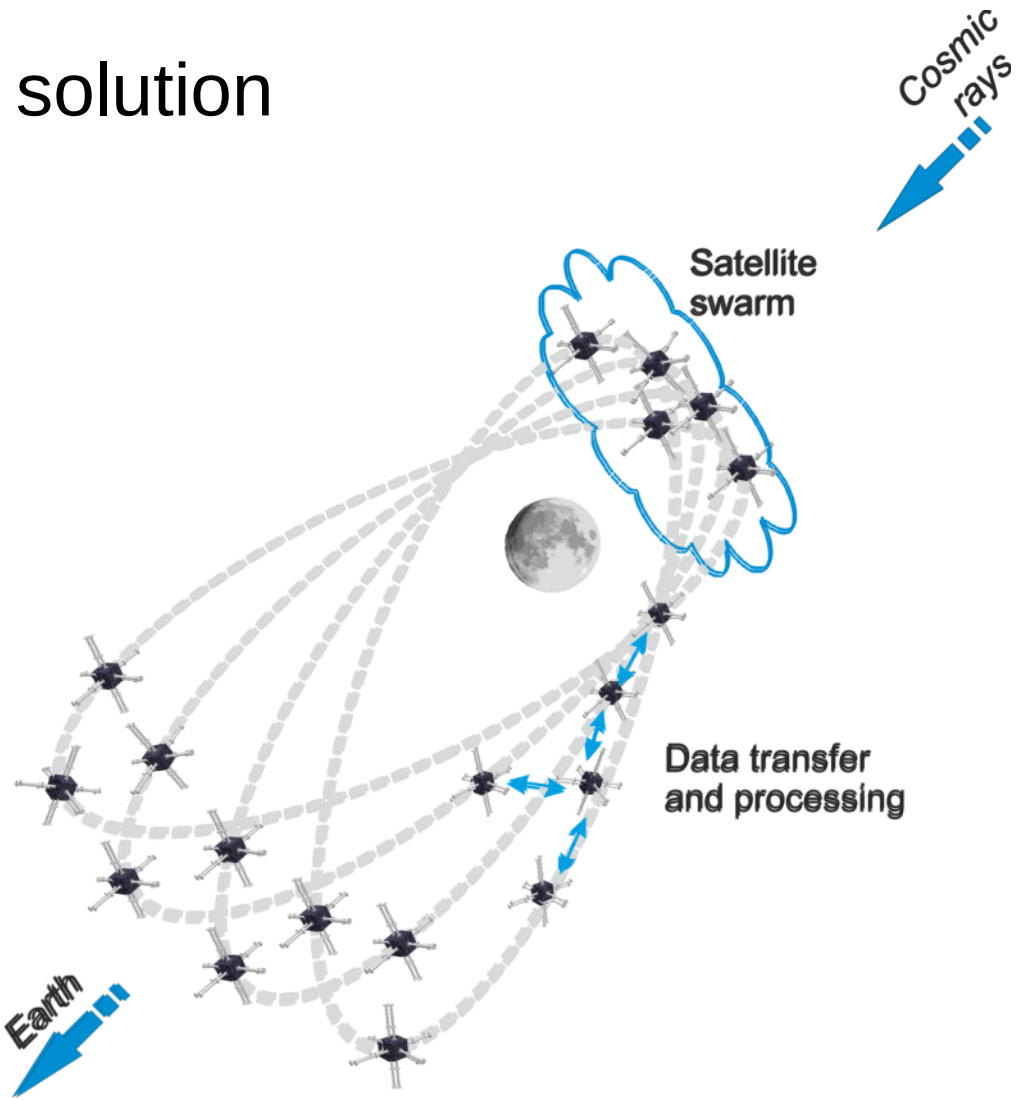


- Nano/small satellites
- Swarm
- Deployable antenna for the frequency band between 1 and 30 MHz
- Ultra-low power receivers
- Intra-satellite communication
- Autonomous distributed processing
- Using diversity techniques for downlink

OLFAR concept

- Efficient and robust solution
  - Low frequencies
    ➔ long wavelength
    ➔ large aperture
  - Feasible

- Swarm advantages
  - Robustness
  - Redundancy
  - Autonomy



OLFAR- requirements

Parameter	Symbol	Range
Frequency range	f	0.1-30 MHz
Sensitivity	S_{min}	< 10 Jy in 1 hour; < 64 mJy in 1 Y
Distance, maximum baselines	D	100 km
Angular resolution	Δα	1 arcminute
Spectral resolution	df	1 kHz
Bandwidth (fractional bandwidth)	Δf	10 MHz
Snapshot integration time	Δt	≥ 1 s
Survey integration time	ΔT	≥ 1 year, nominal 3 years
Number of satellites	Nsat	>50
Range accuracy	ΔD	0.1 λ

Some results

- Team:
  - Four PhD students
  - One Postdoc
  - Lot’s of MSc students
  - University of Twente / Delft University of Technology / ASTRON

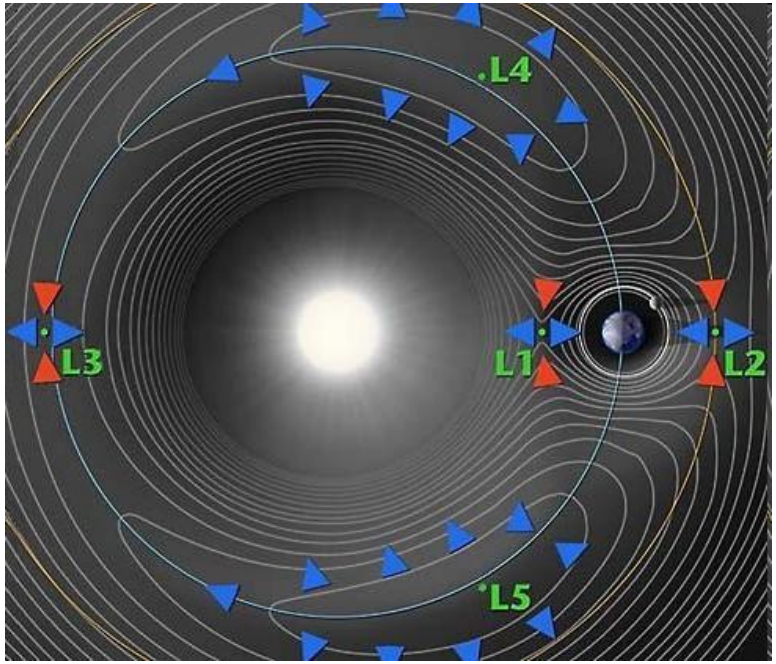
- Topics (selection)
  - Mission analysis
  - Science antenna
  - Intersatellite link
  - ….

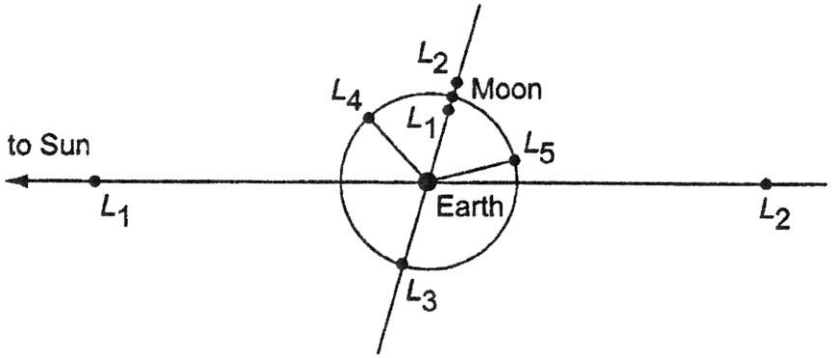
Mission analysis (orbits)

- **Design considerations**
- Astronomical science
- RFI from Earth
- Intergration time, range rate
- Constellation control
  (abs. and rel. position)
- Downlink to Earth

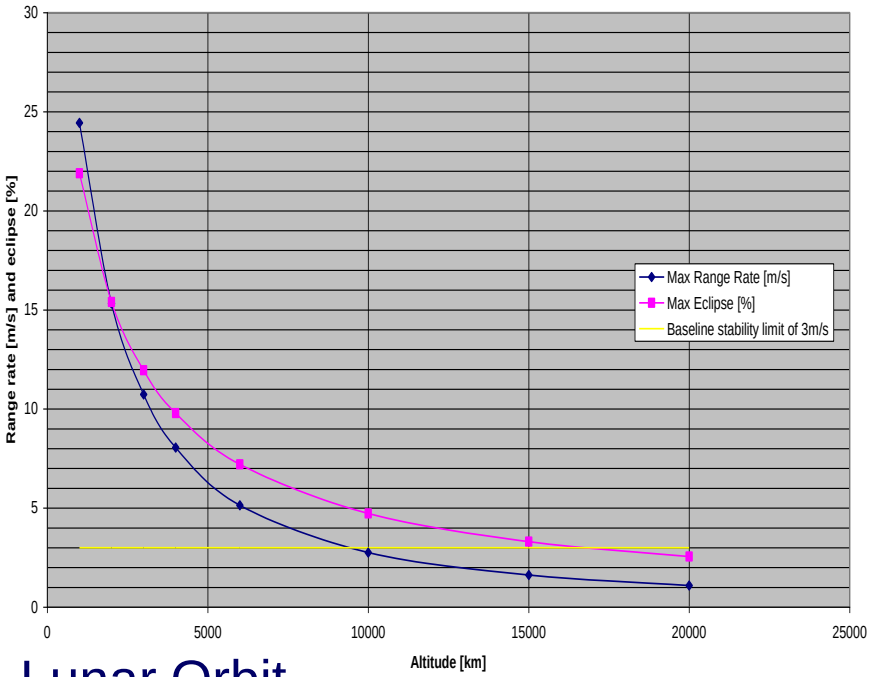
Possible orbits:

- Earth orbit
- **Moon orbit**
- Earth-Moon L2 (saddle point)
- Sun-Earth L4/L5
- Sun-Earth leading/trailing orbit
- **Dynamic solar orbit**
- Sun-Earth L2 (saddle point)
- Moon surface - far side





Mission analysis (orbits)



Lunar Orbit

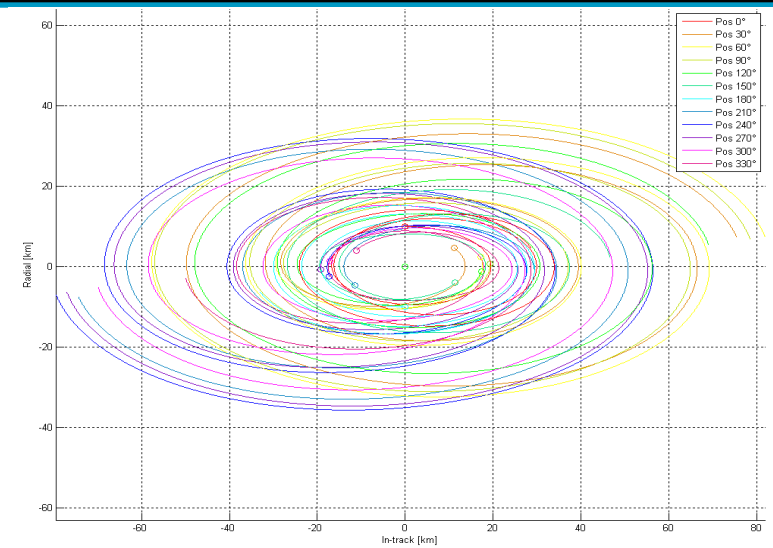
A circular orbit in the lunar equatorial plane

Advantages:

- Stable reference orbit
- Stable relative orbit

Disadvantages:

- Trade-off between eclipse time and range rates



Solar Orbit

An elliptical orbit at a distance of 4-10 million km to Earth

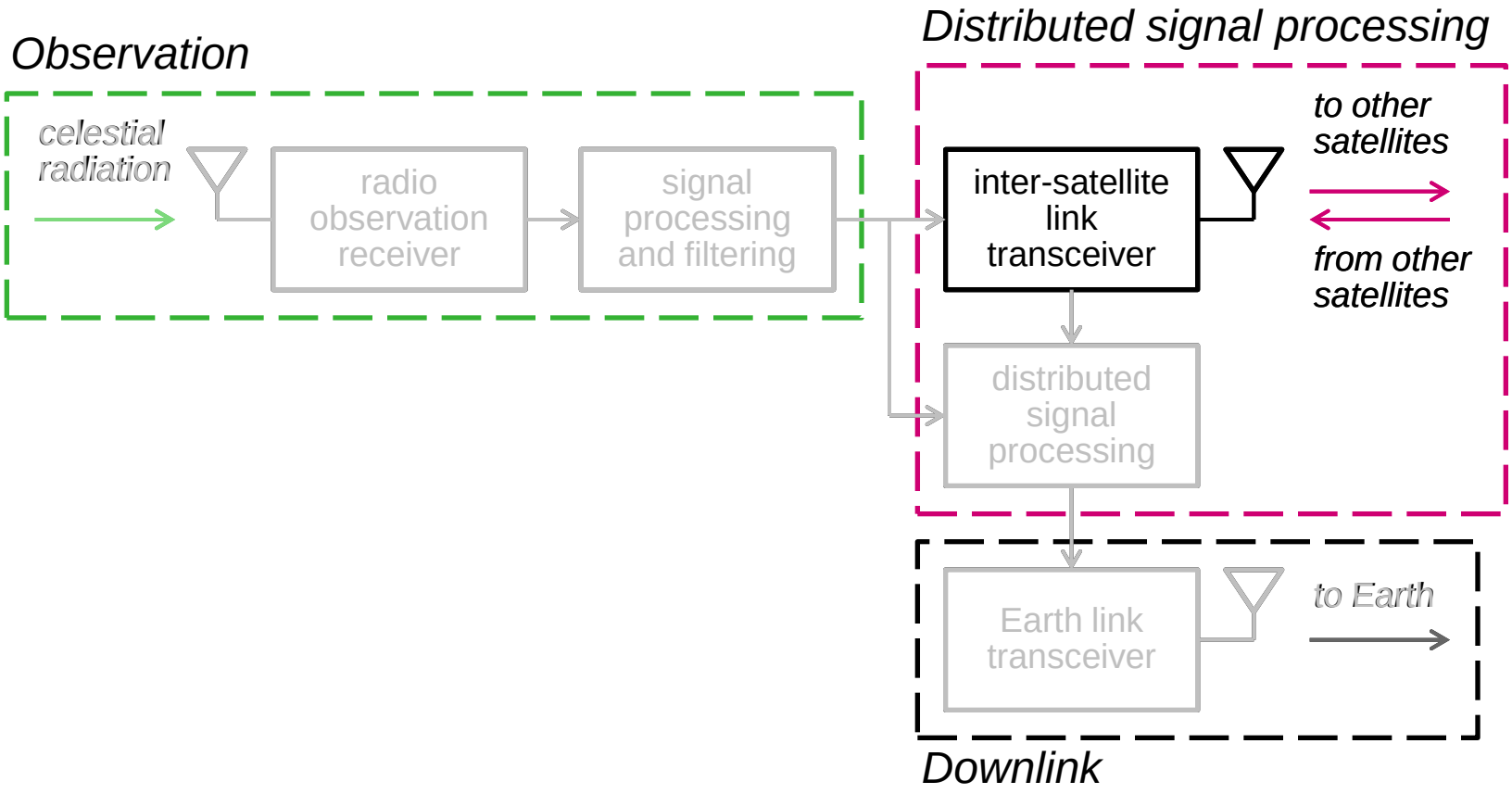
Advantages:

- Very low range rates
- Continuous observations possible
- Stable reference orbit over a very long time

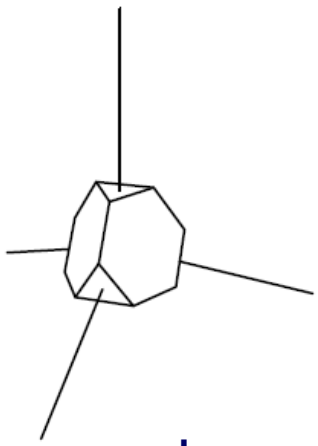
Disadvantages:

- Continuous drift of the satellites, so limited life time
- Very sensitive for velocity changes

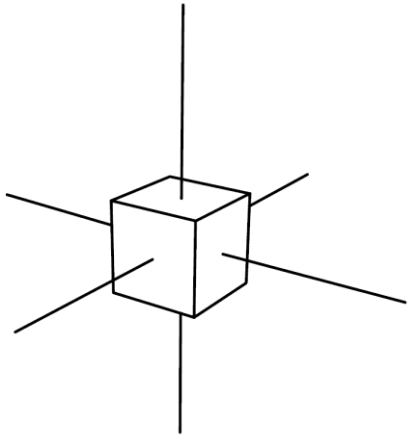
Architecture



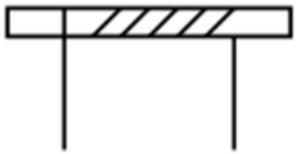
Science antenna



four monopoles on tetraeder



“tripole”



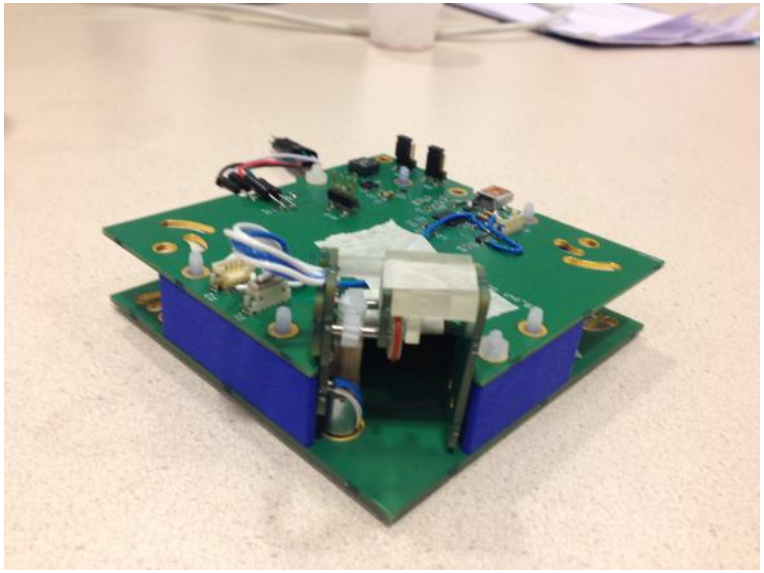
magnetic loop antenna

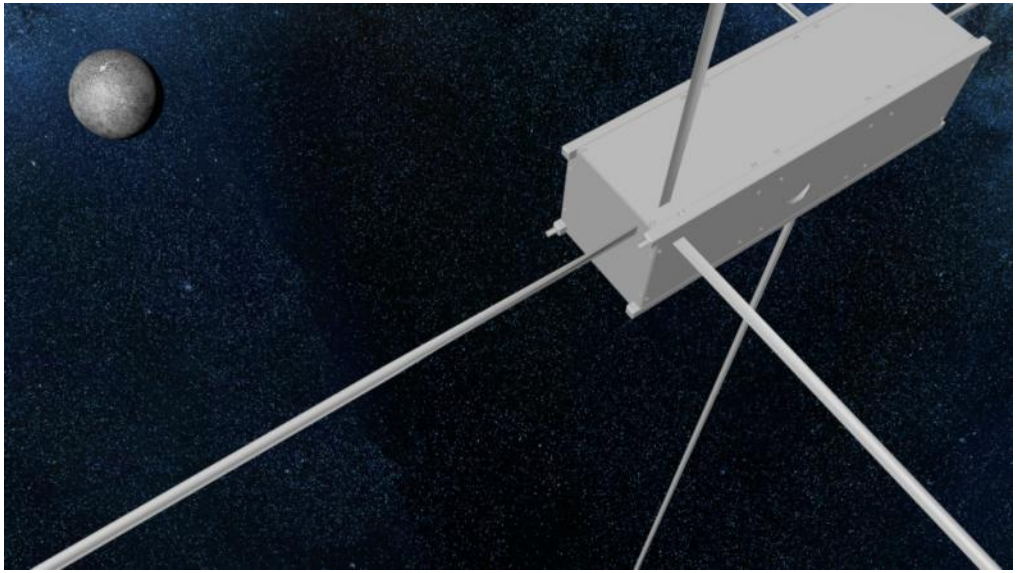
Requirements:

- system needs to be sky noise limited
- Trec<0.1Tsky
- operational bandwidth 1-10 MHz
- instantaneous bandwidth at least 1 MHz
- Small form factor
- Lightweight

Science antenna

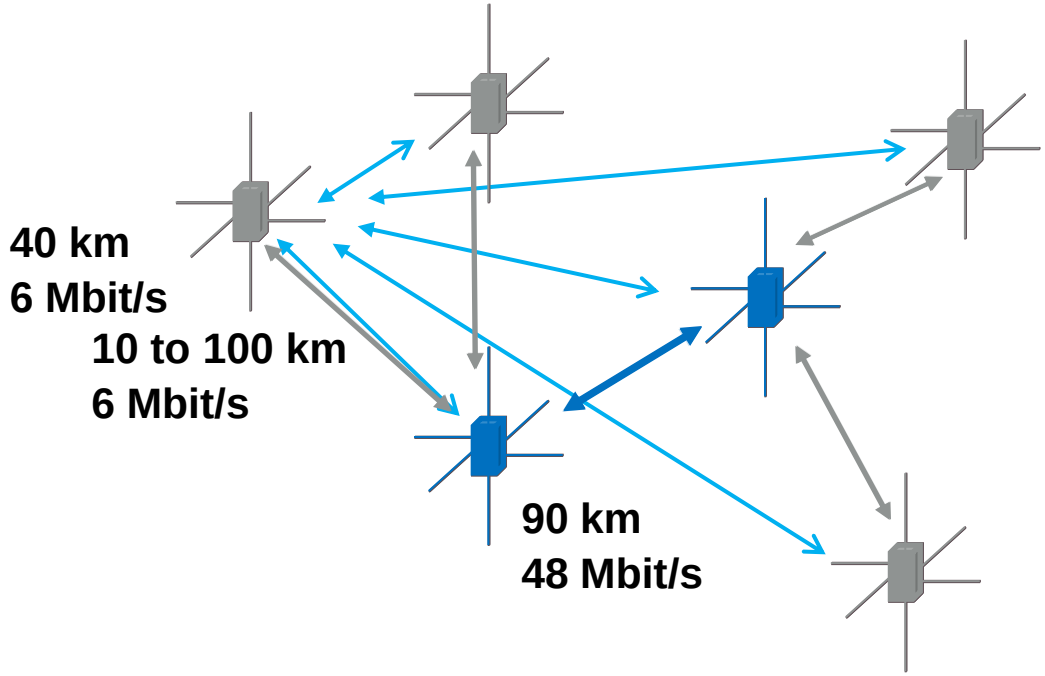
- 3 orthogonally placed dipoles consisting of 6 monopoles, each with a length of 4.9m
- 2 units





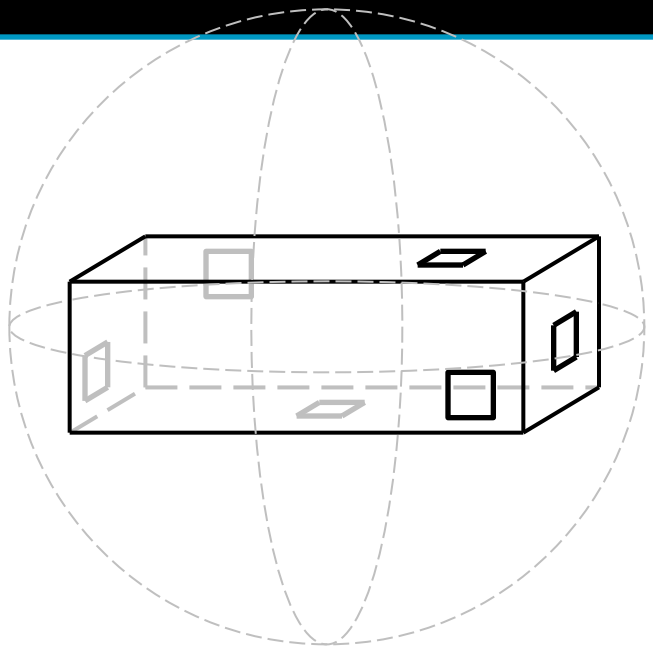
Inter satellite link

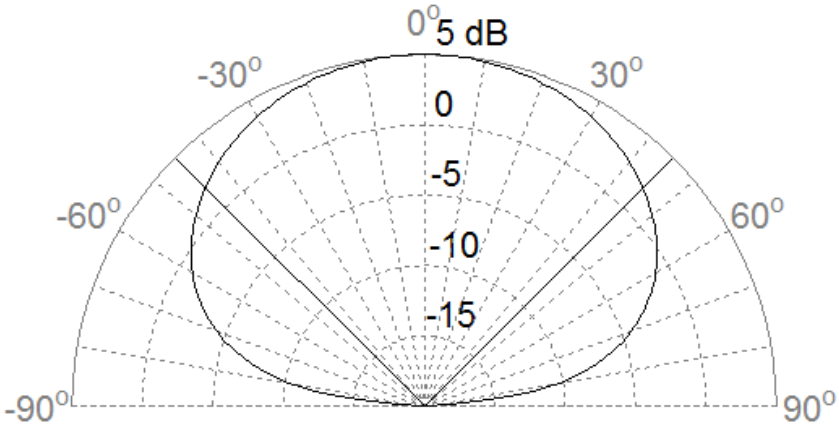
- Adaptive clustering for satellite swarms
  - Hierarchical topology
  - Power efficiency



Antenna configuration

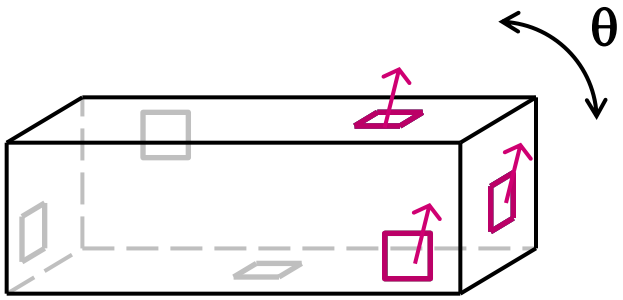
- 4π sr range
- Minimum number of antennas
- One antenna per face
- Radiation pattern
  - 90° 3-dB beamwidth
  - 5 dBi gain





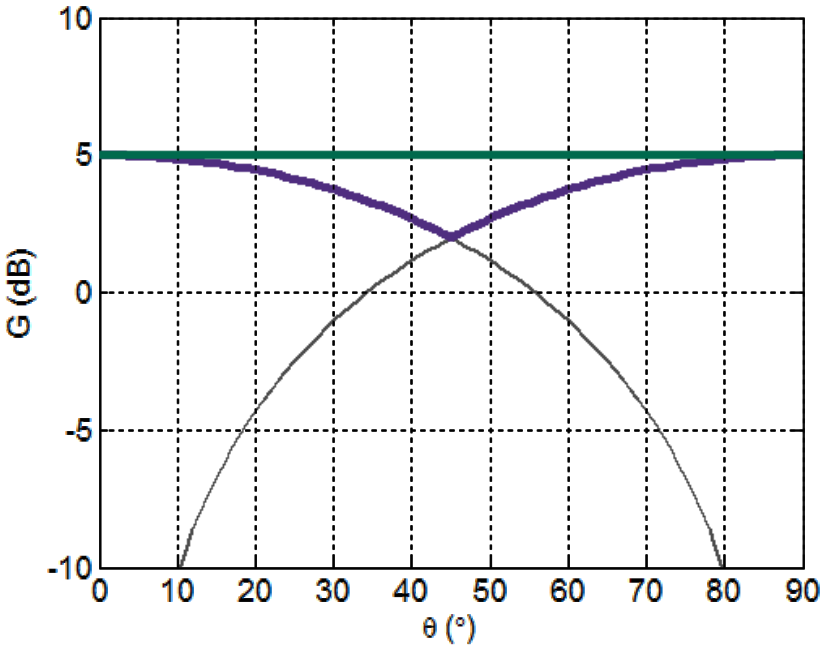
Antenna system control

- Smart antenna system
  - Selection
  - Beamforming

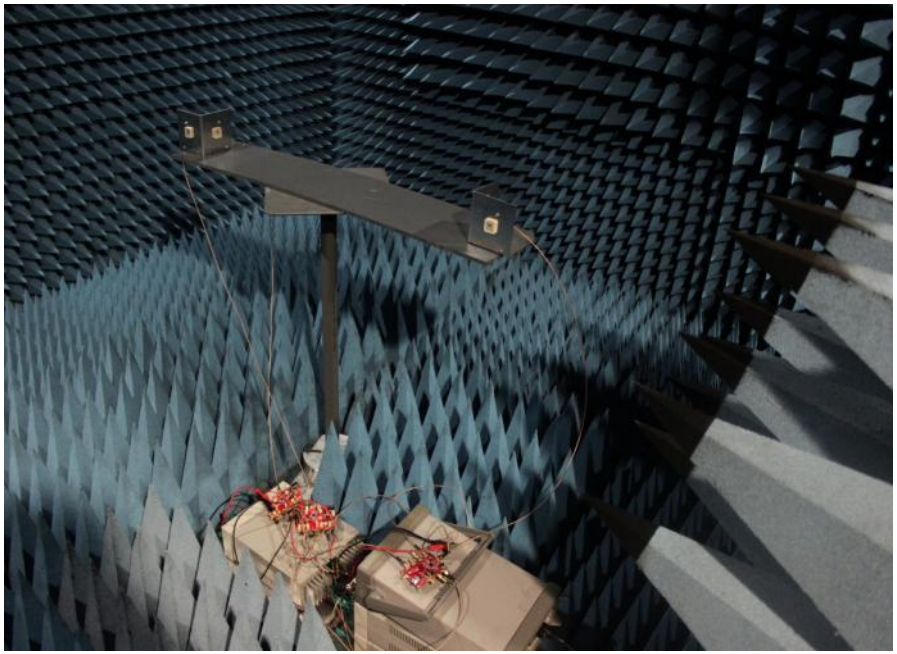


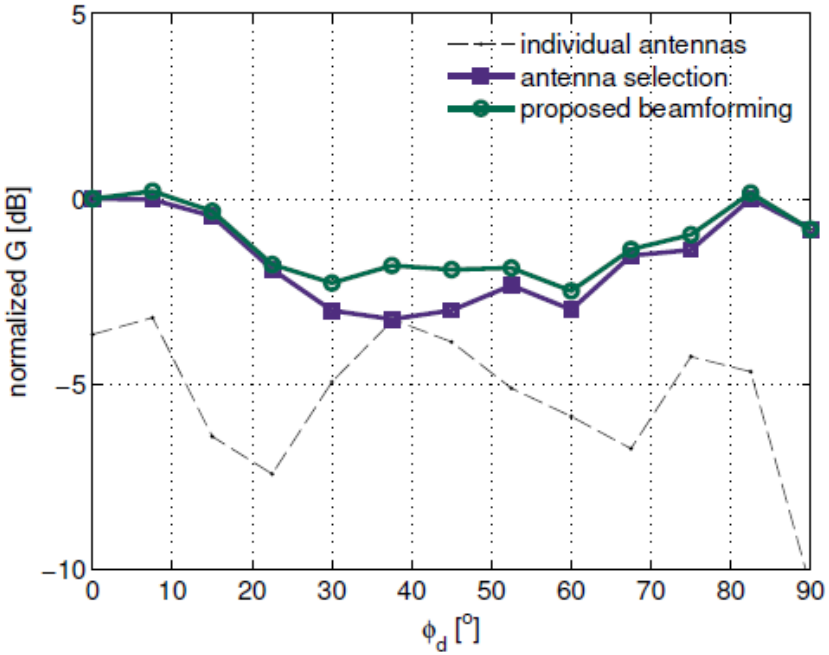
- 3 antennas for each direction

- Maximal gain for any direction
- DOA estimation
- Link tracking



Experiments





Current OLFAR research topics

- Down link
- Solar arrays with integrated phased array transceivers
- Propulsion (see “EPFL - A Couple Drops of Fuel to Get to the Moon with MicroThrust” - http://www.youtube.com/watch?v=YJlSI_l5g4M)
- System design
- On Board Computer with swarm control algorithms
- High accuracy clock/timing
- Clock synchronisation and ranging algorithms
- Antenna deployment mechanisms
- Low power, high bandwidth high sensitivity receiver (i.e. payload package)
- Navigation systems (either pulsar based, or sun/star-sensor based, perhaps using optical navigation techniques)
- Attitude determination systems (star trackers, fine sun-sensors)
- Imaging
- Calibration

Conclusions

- Ultra long wavelength astronomy below 30 MHz is the last unexplored frequency range.
- Due to ionospheric and RFI reasons the best place to be is in space.
- Today’s technology makes a satellite array (swarm) possible.
- A novel antenna system is presented.
- Inter satellite communication is perhaps the most challenging part of OLFAR.

Partners

ASTRON

UNIVERSITEIT TWENTE.

TU Delft
Technische Universiteit Delft

Radboud Universiteit Nijmegen

Dutch Space
an EADS Astrium company

ASTRIUM
AN EADS COMPANY

ISIS

AEMICS

National Semiconductor

AXIOM IC
TWENTE

SystematIC
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esa

BRADFORD
engineering

NLR